

Chapter 13

Numerals and their kin: The view from Slavic

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In this paper, I focus on the meaning of numerals, i.e., lexical items that express a relationship with numbers. I discuss different functions of numerals including their quantifying and arithmetical use and compare various attempts to capture the relationship between those functions. Furthermore, I examine several types of complex numerical expressions such as ordinals, multiplicatives, taxonomic and group numerals in Slavic and beyond. Finally, I investigate data concerning Slavic label numerals, i.e., expressions used to identify entities via association with a number and propose a morpho-semantic analysis of such expressions. The core of the proposal is that the arithmetical meaning is the numerals' underlying semantic core from which both the quantifying and the label meaning are derived.

1 Introduction

Numerals are funny words. On the one hand, it seems that they share some core component that justifies grouping them together as if they formed a single category, a practice common in both descriptive and theoretic approaches to language.¹ On the other hand, though, morphologically and syntactically they form a very heterogeneous class of expressions with different items often exhibiting

¹This intuition dates back at least to the Neo-Latin grammarian tradition, which distinguished between three subclasses within the category *nomen*, i.e., *nomen substantivum*, *nomen adjectivum* and *nomen numerale*. At the same time, different linguistic traditions, including descriptive grammars of Slavic, have proposed conflicting classifications of what counts as a numeral; thus, ...*and their kin* in the title of this paper.



distinct properties. This is the case not only when one compares various subclasses of numerals, e.g., cardinals, ordinals and multiplicatives, but also often within the subclass of cardinals different items have adjectival, nominal, or both nominal and adjectival features while in some languages they seem to behave as verbs (Donohue 2005, Ionin & Matushansky 2018).

In generative linguistics, a lot of work has been focused on establishing the syntactic status of numerals, with different approaches differing in whether they are lexical or functional categories or whether they are heads or maximal projections. These questions have been studied thoroughly with respect to Slavic numerals, with their well-known idiosyncratic properties (e.g., Pesetsky 1982, Babby 1987, Franks 1994, Rutkowski 2002, Klockmann 2012).

On the other hand, since the early days of analytic philosophy and formal semantics much attention has been dedicated to capturing numerals' meaning. At least since Frege (1884), it has been recognized that this is far from trivial. Intuitively, it seems straightforward that numerals are linguistic expressions that somehow describe a numeric quantity. However, beyond this vague characterization it is much less obvious how to state exactly what they actually are.

In this paper, I will discuss the main formal approaches to the meaning of numerals. In doing so, I will focus on two aspects that only recently attracted due attention in the literature. Specifically, I will investigate different uses of basic numerals such as *five* and the behavior of various types of derivationally complex numerical expressions in Slavic. The two sets of data indicate that basic numerals and numerical expressions in general are semantically complex. I will argue that a proper approach to their meaning should describe a compositional mechanism deriving different functions from the underlying arithmetical meaning.

The paper is outlined as follows. In Section 2, I discuss various functions of basic numerals. Section 3 provides an overview of semantic approaches to their quantifying function, as in *five cats*. Section 4 explores the relationship between the quantifying meaning and the ability of numerals to refer to numbers as abstract arithmetical concepts. In Section 5, I review the literature on Slavic complex numerical expressions which presents evidence calling for a compositional treatment. Section 6 is a case study exploring the so-far neglected label use of numerals. Section 7 concludes the paper.

2 Basic numerals and their various flavors

Let us start with basic numerals like English *five*. One of the challenges in capturing their meaning is that they can be used in very different ways (Bultinck 2005,

Geurts 2006). Consider (1). It turns out that a simple word such as *five* is in fact quite tricky. Of course, it can be used to count entities denoted by the modified NP when it appears preminally (1a) but that is not the only function it can have. When it combines with a measure word (1b), it designates a quantity of a substance rather than a plurality of objects. It can also appear in a predicative context (1c) or as part of a measure phrase (1d). And that is not it since it can also be used to refer to an abstract arithmetical concept (1e) or to label an entity (1f).

- (1) a. Five cats meowed.
- b. Five liters of milk got spilled.
- c. These are five cats.
- d. That cat is five years old.
- e. Five is a Fibonacci number.
- f. Player number five twisted her ankle.

Below, I will briefly discuss how the measure, predicative and arithmetical function of numerals relate to their most deeply studied quantifying use. A thorough investigation into properties of the label function will be undertaken in Section 6.

2.1 Quantifying vs. measure use

The mainstream research has been primarily focused on the quantifying use of numerals (1a). This seems justified since counting objects is perhaps the first thing that comes to mind when we think about expressions such as *five*. However, when a numeral appears in a measure phrase (1b), it seems to involve measuring within a certain dimension, e.g., volume, rather than counting objects.

Intuitively, the difference between counting and measuring is that the former is about specifying how many discrete objects of a certain kind there are, whereas the latter determines some quantity in relevant measure units. Admittedly, some proposals attempt to reduce measuring to a particular type of counting based on the individuation in terms of quantity (Lyons 1977, Matushansky & Zwarts 2017). Consequently, measuring would simply be counting units determined by measure words. An opposing approach treats counting as a form of measuring, i.e., measuring a quantity of a plural individual in terms of natural or object units (Krifka 1989, 1995). Yet, there is also the third account which views counting and measuring as two independent operations (Rothstein 2017).

The contrast between the two functions becomes clear when one considers classifier constructions involving container nouns (2). Such expressions are ambiguous between a measure (content) and counting (container) interpretation

(e.g., Landman 2004, Grimshaw 2007, Partee & Borschev 2012). On the measure reading of (2a), Tomek flushed wine to the quantity of two bottles. On the counting reading, on the other hand, he did something quite spectacular since two actual bottles containing wine went down. Moreover, derivational morphology seems to be sensitive to the distinction since the suffix *-ful* selects only for the measure sense of a container noun (2b)–(2c) (Rothstein 2011).

- (2) a. Tomek flushed two bottles of wine through the toilet.
- b. Romek added two { glasses / glassfuls } of wine to the soup.
- c. Marek brought two { glasses / #glassfuls } of wine for our guests.

Another argument for distinguishing counting and measuring as two distinct operations comes from distinct restrictions on the domain of quantification (Wągiel 2018). Though both measuring and counting quantify over entities that need to be disjoint (Landman 2016), only the latter requires objects individuated as coherent integrated wholes. To realize the contrast, imagine someone has spilled some wine on the table in such a way that there are two separate blobs *a* and *b* whose volume is exactly one and a half milliliters each. In such a scenario, (3a) is true despite the fact that one of the three milliliters must be split between a portion of *a* and a portion of *b*. On the other hand, (3b) is simply false. This contrast shows that measuring, unlike counting, ignores individuation in terms of integrity.

- (3) SCENARIO: There are exactly two 1.5 ml blobs of wine on the table.
- a. There are three milliliters of wine on the table. TRUE
- b. There are three objects on the table. FALSE

Though monotonic systems of measurement track certain part-whole relations (Schwarzschild 2002), they are not sensitive to the spatial arrangement of parts making up a whole. This suggests that despite sharing a common core counting and measuring are two distinct things.

2.2 Quantifying vs. predicative use

It is well known that cardinals used as quantifiers have both lower and doubly bounded readings.² For instance, *five cats* in (4a) allows for an *at least* reading, as witnessed by the compatibility with the *in fact* clause. Similarly, (4b) is typically interpreted in a way that Tomek must take at least three cards.

²For an overview of the literature on numerals and scalar implicatures, see, e.g., Spector (2013).

- (4) a. Five cats live in the barn; in fact, six cats live there.
- b. Tomek must take three cards.

However, numeral NPs in predicate position systematically lack a lower-bounded interpretation (Partee 1987, Landman 2003, Geurts 2006). For instance, (5a) can only get the *exactly* reading. This is further corroborated by the infelicity of (5b).

- (5) a. These are five cats.
- b. # The guests are three filmmakers; in fact, they are four filmmakers.

Furthermore, it was observed that also bare cardinals can appear as predicates in predicate position (6) (Rothstein 2017). This resembles the behavior of adjectives rather than determiners.

- (6) a. The apostles are twelve.
- b. The planets are eight.

Similar to numeral NPs, bare cardinals in predicate position lack an *at least* reading. Thus, (6a) and (6b) are true if there are exactly twelve apostles and exactly eight planets, respectively. Moreover, sentences such as (7) are infelicitous.

- (7) # My reasons for saying this are four; in fact they are five.

The contrasts discussed above indicate that the meaning of the predicative use of cardinals differs from their semantics in the quantifying function.

2.3 Quantifying vs. arithmetical use

It is not always the case that numerals designate a plurality or measure. In (1e), *five* refers to an abstract number concept rather than to a collection of entities. In this use, numerals behave differently than in their quantifying function (Bultinck 2005, Rothstein 2017, Wągiel & Caha 2020). Specifically, arithmetical concepts can have special properties such as being prime (8a) and they can occur in mathematical statements (8b) and dedicated grammatical constructions (8c). Finally, when compared, the dimension of comparison is based on their relative ordering rather than, e.g., on the physical size (8d).

- (8) a. Five is prime.
- b. Ten divided by five equals two.
- c. Hanna can count up to five.
- d. Five is bigger than four.

This behavior contrasts with quantifying numerals. (9a) is odd since being prime is not a property that can be attributed to a collection of things. Similarly, expressions denoting a plurality of entities are illicit in constructions calling for a numeric value such as (9b)–(9c) and (9d) has different truth conditions than (8d), e.g., it would not be true if one compared five cherries with four watermelons.

- (9) a. # Five things are prime.
- b. # Ten things divided by five things equals two things.
- c. # Hanna can count up to five things.
- d. # Five things are bigger than four things.

Furthermore, numerals referring to number concepts are incompatible with modifiers such as *more than* and *at least* (10a). Finally, the arithmetical function is always associated with an *exactly* reading (10b).

- (10) a. # More than five is prime.
- b. # Two multiplied by five equals ten, if not more.

As we saw, numerals can be used in various ways each of which has different semantic characteristics. In the next two sections, I will review major approaches to their quantifying meaning and compare how they account for at least some aspects of the flexibility discussed above.

3 Theories of cardinals

Given the variety of uses discussed in the previous section, any quest for *the* meaning of cardinal numerals is probably a misunderstanding. Rather, a theory of cardinals should be able to capture the relationship between meanings of cardinals in their various functions. And yet, the mainstream research has been mainly focused on the quantifying use often ignoring how it relates to other uses (with Bylinina & Nouwen's 2020 systematic inquiry being a notable exception).

3.1 Cardinals as determiners

Let us start with the earliest and most prominent account of the meaning of cardinal numerals, namely the Generalized Quantifiers (GQ) approach (Barwise & Cooper 1981; though it can be traced back via Montague to Frege). The intuition behind this analysis is that *five* is in fact a determiner similar to *some*, *most* or *all*, as suggested by its occurrence in prenominal position (11).

- (11) { Some / Most / All / Five } cats meowed.

In the GQ theory, a determiner expresses a particular relation holding between two sets (type $\langle\langle e, t \rangle, \langle\langle e, t \rangle, t \rangle\rangle$), i.e., a set denoted by the NP and a set denoted by the VP. In (11), *some* yields the truth value True if the intersection between the set of cats and the set of entities that meowed is non-empty (12a). Similarly, *five* returns True if the cardinality of the set of cats that meowed equals 5 (12b).

- (12) a. $\llbracket \text{some} \rrbracket = \lambda P \lambda Q [P \cap Q \neq \emptyset]$
 b. $\llbracket \text{five} \rrbracket = \lambda P \lambda Q [|P \cap Q| = 5]$

The GQ approach aims to develop a unified analysis of all quantified DPs. Further, it can be enriched with a type-shifting mechanism which allows for systematic mapping between different semantic types (Partee 1987). Consequently, the theory can account for predicative uses of numeral NPs like (1c) by lowering the type of a general quantifier ($\langle\langle e, t \rangle, t \rangle$) to the type of a predicate ($\langle e, t \rangle$).

However, for some time it has been realized that the GQ approach is most probably not a good way of capturing what cardinals are (e.g., Krifka 1999, Landman 2003). In the next sections, I will discuss the well-known problems with the GQ account and main alternative approaches.

3.2 Cardinals as predicates

There are several problems with the GQ approach to cardinals. First of all, there is a well-known asymmetry between DPs with numerals and DPs with regular determiners in predicate position. For instance, DPs headed by *every* cannot be used predicatively (13a). Similarly, (13b) is infelicitous on a non-partitive interpretation. Given the felicity of (1c), repeated here as (13c), and other sentences discussed in Section 2.2, this fact is puzzling (Landman 2003).

- (13) a. # Hanna is every filmmaker at the party.
 b. # The guests are { most / some } filmmakers.
 c. These are five cats.

Furthermore, (14a) shows that bare determiners cannot be used predicatively. This contrasts with (6), in which the cardinal patterns with adjectives (14b). Since there is no standard type-shifting rule allowing for mapping the type of determiners ($\langle\langle e, t \rangle, \langle\langle e, t \rangle, t \rangle\rangle$) onto the type of predicates ($\langle e, t \rangle$), the predicative use of bare cardinals is unaccounted for. But even if such a rule were postulated, it would still fail to explain the contrast between cardinals and determiners.

- (14) a. # My reasons for saying this are { most / all / every }.
 b. My reasons for saying this are { serious / personal / five }.

The last problem concerns the fact that cardinals can appear along with bona fide determiners within a single DP (e.g., Rothstein 2017). From a GQ perspective, the compatibility of *five* with the definite article (15a) and *every* (15b) is unexpected and raises questions regarding the semantic contribution of each of the elements.

- (15) a. The five cats that I saw meowed.
 b. Every two students got to share a hotel room.

The evidence discussed above motivated an alternative approach, which treats cardinals as predicates (Landman 2003, 2004). Within this framework, cardinals get an interpretation very similar to that of intersective adjectives, specifically they express a cardinality property. For this approach to work, it is required to distinguish between two types of entities within the domain of individuals, namely atoms, i.e., singular entities, and pluralities (Link 1983). While atoms are minimal elements of a nominal denotation, pluralities are formed from atoms via a pluralization operation $*$ which closes the predicate under sum, i.e., takes a set of atomic entities and returns that set extended with all the pluralities that can be formed by summing the atoms. In addition, the two types of individuals are associated with each other via a part-whole relation defined in terms of mereological parthood, e.g., an atomic entity a is part of a plurality $a \oplus b$.³

According to the semantics in (16a), *five* denotes a set of pluralities each of which contains 5 atomic entities. Similar to intersective adjectives, cardinals combine with NPs they modify via Predicate Modification (Heim & Kratzer 1998). Therefore, when *five* is composed with *cats* (16b), the resulting phrase denotes the intersection of a set of pluralities of cats and a set of plural individuals that consist of five atomic entities, i.e., a set of sums of five cats.

- (16) a. $\llbracket \text{five} \rrbracket = \lambda x [\#(x) = 5]$
 b. $\llbracket \text{five cats} \rrbracket = \lambda x [*CAT(x) \wedge \#(x) = 5]$

The approach discussed above gives a straightforward explanation for the contrasts between cardinals and determiners like *every* and *most* in predicate position as well as the felicity of constructions such as (15). However, it faces some challenges as well.

³For an introduction to mereological theories of plurality, see, e.g., Champollion & Krifka (2016).

3.3 Cardinals as predicate modifiers

A problem with treating cardinals as cardinal properties is that cross-linguistically the use of bare cardinals in predicate position is very restricted (Ionin & Matushansky 2018: 33–34). For instance, in Russian (17) is ungrammatical. Similarly, Dutch (18) can only mean that the children are two years old and not that they are two in number. And even in English the predicative use of bare cardinals is heavily restrained, as witnessed by the fact that examples such as (19) are highly degraded. If cardinals are assigned type $\langle e, t \rangle$, then this is rather startling.

- (17) * *Deti byli { dva / dvoe }.*
 children were two two.GNDR
 Intended: ‘The children were two in number.’ (Russian)

- (18) # *De kinderen zijn twee.*
 the children are two
 Intended: ‘The children are two in number.’ (Dutch)

- (19) ?? The chairs in this room are twelve.

Another issue concerns the fact that what counts as ‘one’ seems to be highly dependent on the meaning of the modified NP. Arguably, atoms should be defined relative to a particular property rather than in absolute terms. To see why, let us consider partitive constructions (20) (see Chierchia 2010, Wągiel 2018). Here, (20a) is weird since the numeral NP seems to designate a pair of body parts that disappeared, i.e., it is not referring to two subsets of boys. Yet, in (20b) quantification over subgroups of boys is possible. Importantly, neither *parts of the boys* nor *parts of the group* denote entities that are atomic in any absolute sense since each can consist of parts that themselves fall into the extension of the corresponding phrase. This suggests that the cardinal determines what counts as ‘one’ relative to the denotation of the modified NP and it is not obvious how this fact could be captured if cardinals were interpreted simply as intersective predicates.

- (20) SCENARIO: While hiking at night, a group of boys divided into five teams to explore a vast area but only three teams arrived at the meeting point.
 a. # Two parts of the boys went missing.
 b. Two parts of the group went missing.

An alternative to treating cardinals as cardinal properties is to analyze them as predicate modifiers (type $\langle \langle e, t \rangle, \langle e, t \rangle \rangle$) equipped with a pluralization operation *

and a measure function $\#(P)$ (Krifka 1989, 1995). In (21a), $*$ closes the predicate under sum, i.e., takes a set of atomic entities and returns that set extended with all the pluralities one can form by summing atoms. On the other hand, the operation $\#$ returns for a property P a measure function that yields pluralities of five individuals having that property. In other words, *five* maps pluralities of entities onto the natural number 5.⁴ When the cardinal combines with an NP, we obtain a set of relevant plural individuals (type $\langle e, t \rangle$). For instance, *five cats* denotes a set of pluralities of cats each of which consists of five cats (21b). Thus, what counts as ‘one’ in this system is relativized to the denotation of the NP, and this together with some additional assumptions allows for capturing partitives like (20) where reference to absolute atoms is problematic (see also Chierchia 2010, Wągiel 2018).

- (21) a. $\llbracket \text{five} \rrbracket = \lambda P_{\langle e, t \rangle} \lambda x_e [*P(x) \wedge \#(P)(x) = 5]$
 b. $\llbracket \text{five cats} \rrbracket = \lambda x_e [*CAT(x) \wedge \#(CAT)(x) = 5]$

Another option is to treat cardinals as predicate modifiers providing the cardinality of a partition of a plural entity (Ionin & Matushansky 2006, 2018). A partition of a plurality x is a cover of x , i.e., a set of entities such that the sum of all those entities forms x ; in addition, it is a cover whose cells do not overlap, i.e., do not share any parts (see Gillon 1987, Schwarzschild 1996). In (22a), S is a partition Π of an individual x and the cardinality of S equals 5. The cardinal combines with the NP via Function Application and the result is of type $\langle e, t \rangle$. For instance, *five cats* gets the meaning in (22b), i.e., it denotes a set of pluralities divisible into 5 non-overlapping entities each of which has a property of being a cat.⁵

- (22) a. $\llbracket \text{five} \rrbracket = \lambda P_{\langle e, t \rangle} \lambda x_e \exists S_{\langle e, t \rangle} [\Pi(S)(x) \wedge |S| = 5 \wedge \forall s \in S [P(s)]]$
 b. $\llbracket \text{five cats} \rrbracket = \lambda x_e \exists S_{\langle e, t \rangle} [\Pi(S)(x) \wedge |S| = 5 \wedge \forall s \in S [\llbracket \text{cat} \rrbracket(s)]]$

Notice, however, that both approaches discussed above require the denotation of NPs the cardinal combines with to be singular, i.e., to consist only of atoms. It is postulated that the source of plurality is the numeral and the plural marking on the noun is, e.g., due to agreement with no semantic interpretation. Supporting evidence comes from expressions such as (23) where the plural is not associated with a plurality (Krifka 1989, Bylinina & Nouwen 2018) and from languages like Finnish and Turkish which display the singular/plural distinction but in which unlike in, say, English cardinals systematically require singular NPs (24).

⁴This is a slight simplification since in fact Krifka (1989) postulates a special operation nu for measuring pluralities in terms of ‘natural units’ they consist of, whereas Krifka (1995) proposes ou for measuring the number of ‘object units’ realizing a particular kind.

⁵The system is devised this way to provide a unified analysis of both simple and complex numerals like *five hundred*.

- | | |
|--|---|
| (23) a. zero students
b. 1.0 students | (24) üç { çocuk / *çocuklar }
three child children
'three children' (Turkish) |
|--|---|

Finally, analyzing cardinals as predicate modifiers makes it easier to account for restrictions on bare cardinals in predicate position (18)–(19), by appealing to the well-described peculiarities of the copula which would allow for a type-shift only under particular circumstances (Ionin & Matushansky 2018: 33–34).

3.4 Cardinals as degree quantifiers

The last approach to be discussed here builds on two observations regarding the behavior of cardinals. The first one is that sentences with existential modals (25) are ambiguous (Kennedy 2013, 2015). On the strong reading, (25a) means that Hanna is allowed to eat five cookies and she is not allowed to eat more. The strong reading is probably the most natural interpretation of (25a). There is, however, also a weak reading which merely states that eating five cookies by Hanna is allowed without saying anything about eating other quantities of cookies. This interpretation becomes more prominent in questions (25b).

- (25) a. Hanna is allowed to eat five cookies.
b. Is Hanna allowed to eat five cookies?

The second observation is that numerals can take scope independently of the rest of the NP. The evidence comes from decimals (Kennedy & Stanley 2009). Crucially, (26a) does not entail the existence of families with 2.1 cats. This contrasts with (26b) which is infelicitous due to such an entailment.

- (26) a. The average Polish family has 2.1 cats.
b. # A normal Polish family has 2.1 cats.

This suggests that after all cardinals are quantifiers. We already saw that the analysis of *five* as a determiner is most probably incorrect and clearly treating it as a quantifier over individuals (type $\langle\langle e, t \rangle, t\rangle$) would not make much sense. However, an interesting idea is to interpret cardinals as quantifiers over degrees, i.e., expressions of type $\langle\langle d, t \rangle, t\rangle$ (Kennedy 2013, 2015). Degrees (type *d*) are objects similar to individuals (type *e*) with the crucial difference that the domain of degrees D_d , unlike the domain of individuals D_e , is ordered.⁶ Hence, on the degree

⁶For more discussion and an introduction to degree semantics, see, e.g., Morzycki (2016: Ch. 3).

quantifier analysis *five* denotes a set of degree properties whose maximal value equals 5 (27a). A sentence with a numeral NP is then interpreted as in (27b), e.g., the maximal number of entities that Hanna had and that are cats is 5.

- (27) a. $\llbracket \text{five} \rrbracket = \lambda D_{\langle d,t \rangle} [\text{MAX}(D) = 5]$
 b. $\llbracket \text{Hanna had five cats} \rrbracket =$
 $= \text{MAX}\{n \mid \exists x [\text{HAD}(x)(\text{HANNA}) \wedge *_{\text{CAT}}(x) \wedge \#(x) = n]\} = 5$

The analysis of cardinals as degree quantifiers provides a promising perspective to explain interactions between modals and modified cardinals such as *more than five* and *at least five* (Nouwen 2010). In the next section, I will discuss the arithmetical function of numerals and how it relates to the theories described so far.

4 Relating cardinalities and number concepts

The theories of numerals discussed in the previous section focus on the quantifying function. However, we already saw in Section 2.3 that numerals can also be used to refer to abstract number concepts.

4.1 Numerals as names of numbers

In the arithmetical function, numerals seem to behave as proper names of number concepts (type n) or, alternatively, degrees (type d).⁷ From this perspective, the arithmetical meaning of *five* is simply the number 5 (28a). Expressions such as *prime* would then denote properties of numbers (type $\langle n, t \rangle$), e.g., the extension of (28b) would be the set $\{2, 3, 5, 7, 11, 13, \dots\}$. On the other hand, expressions used to formulate mathematical statements such as *plus*, *times*, *divided by* and *equals* seem to describe relations between number concepts. For instance, (28c) is interpreted in terms of addition of two numeric values.

- (28) a. $\llbracket \text{five} \rrbracket = 5$
 b. $\llbracket \text{prime} \rrbracket = \lambda n_n [\text{PRIME}(n)]$
 c. $\llbracket \text{plus} \rrbracket = \lambda n_n \lambda m_n [n + m]$

The key question concerns the relationship between names of numbers, i.e., the arithmetical use, and meanings of cardinals used in the quantifying function. In the next sections, I will discuss two possible approaches to this issue.

⁷I assume here that there is no difference between the two notions and following the widespread convention in the literature on the arithmetical use of numerals I will simply use the label n .

4.2 Deriving number concepts from cardinalities

The mainstream view is to take the quantifying meaning as basic, be it the predicate, predicate modifier or degree quantifier analysis, and to derive the arithmetical meaning from it. Let us discuss three variants of that view.

As we already saw in Section 3.2, on the predicate analysis cardinals denote a cardinal property (29a). Rothstein (2017) assumes this semantics to be basic.⁸ The arithmetical meaning is then derived via a special type-shifting operation $\cap$ which when applied to a cardinal property yields a proper name of a number (29b).

- (29) a. $\llbracket \text{five} \rrbracket_{\langle e,t \rangle} = \lambda x_e [\#(x) = 5]$
 b. $\llbracket \text{five} \rrbracket_n = \cap \llbracket \text{five} \rrbracket_{\langle e,t \rangle} = 5$

The proposal is motivated by an analogy with adjectival and nominal uses of expressions describing properties such as *white* and *wise* ~ *wisdom*. It builds on Chierchia's (1985) theory of predication according to which every property has an entity correlate derived by $\cap$. For example, the predicate *white* can be turned into the name of the property of being white. Similarly, $\cap$ turns *five* into the name of the property of being five in number. The reverse shift is also possible via $\cup$.

However, as argued by Ionin & Matushansky (2018: 31–33) the above analogy is problematic since an arithmetical statement such as those in Section 2.3 requires an argument to be the number 5 itself rather than the name of the property of being five in number. One can add that though from a philosophical perspective it is unclear what numbers ultimately are, from a linguistic point of view it is crucial that while there are many contexts in which, e.g., *wisdom* and *being wise* seem to be entirely synonymous (30), there are few (if any) contexts in which *five* and *being five in number* are equivalent (31).

- (30) a. Being wise is the art of knowing what to overlook.
 b. Wisdom is the art of knowing what to overlook.
- (31) a. Five is the fourth prime number.
 b. # Being five in number is the fourth prime number.

Consequently, in order to derive the arithmetical function Ionin & Matushansky (2018: 26–27) propose a null nominalizing suffix *NOMNUM*, which applied to a predicate modifier, turns it into a numeric singular term. More specifically, for any

⁸In fact, this is a slight simplification since the theory distinguishes typically between lower numerals and 'lexical powers', i.e., multiplicands such as *hundred* in *five hundred*.

cardinal numeral it yields the corresponding cardinality. An empirical argument for such an account comes from the fact that numerals as names of numbers typically display full morpho-syntactic uniformity suggesting they are derived expressions. On the other hand, cardinals in the quantifying function often show variation with respect to ϕ -features, e.g., grammatical gender.

Another approach is Kennedy's (2015) proposal to derive the arithmetical meaning from the quantifying one by the standard type-shifting operations BE and IOTA defined in (32) (Partee 1987). BE takes a quantifier and returns a property, whereas IOTA yields the unique entity of which the relevant property is true.

- (32) a. $\text{BE} = \lambda \mathcal{P}_{\langle \langle e, t \rangle, t \rangle} \lambda x_e [\mathcal{P}(\lambda y_e [y = x])]$
 b. $\text{IOTA} = \lambda P_{\langle e, t \rangle} \iota x [P(x)]$

Generalizing the type-shifts defined above so that they can also apply to quantifiers over degrees and properties of degrees (or numbers), respectively, allows to derive the arithmetical meaning by applying BE and IOTA consecutively to the degree quantifier semantics. Specifically, when BE applies to a set of degree properties whose maximal value equals 5 (27a), the result in (33a) is the singleton set $\{5\}$. The subsequent application of IOTA yields the number 5 (33b).

- (33) a. $\text{BE}(\llbracket \text{five} \rrbracket) = \lambda n_n [n = 5]$
 b. $\text{IOTA}(\text{BE}(\llbracket \text{five} \rrbracket)) = \text{IOTA}(\lambda n_n [n = 5]) = 5$

An advantageous feature of this approach is that it allows for relating the quantifying and the arithmetical meanings using standard and independently motivated type-shifting machinery. Importantly, however, by doing so it can only derive the arithmetical function from the quantifying one but not vice versa. We will see that this restriction, shared with Ionin & Matushansky's approach, makes a problematic prediction concerning morphological marking. In the next section, I will discuss accounts that derive cardinalities from the arithmetical meaning.

4.3 Deriving cardinalities from number concepts

In opposition to the view that the quantifying function of numerals is basic, a number of approaches build on a contrary intuition, namely that underlyingly numerals are simply names of number concepts (e.g., Scha 1981, Krifka 1989, Scontras 2014, Wągiel 2018). According to that alternative view, cardinals in prenominal position are in fact syntactically and semantically complex

An early theory of cardinals postulating that they are derived from number concepts was developed by Scha (1981). This system distinguishes between numbers, numerals and numerical determiners. The first simply denote number concepts. On the other hand, the NUMERAL head shifts an integer (type n) to a cardinal predicate while the numerical D head transforms such a cardinal predicate into a determiner with a GQ-style semantics (34).⁹

$$(34) \quad [D_{\langle\langle e,t \rangle, \langle\langle e,t \rangle, t \rangle\rangle} [NUMERAL_{\langle e,t \rangle} [NUMBER_n \text{ five}]]]$$

A related idea is the core of the system developed by Hackl (2000) who makes use of a covert quantificational operator MANY to turn a number concept into a determiner (35a). On this approach, the quantifying cardinal is accompanied by an additional null element that ensures the shift (35b). The result combines with the denotation of the NP to form a generalized quantifier.

$$(35) \quad \begin{array}{ll} \text{a. } \llbracket \text{MANY} \rrbracket = \lambda n_n \lambda P_{\langle e,t \rangle} \lambda Q_{\langle e,t \rangle} \exists x_e [\#(x) = n \wedge P(x) \wedge Q(x)] \\ \text{b. } \llbracket \text{five cats} \rrbracket = \llbracket [[5 \text{ MANY}] \text{ cats}] \rrbracket \end{array}$$

Another proposal on how to derive the quantifying meaning from the arithmetical one dates back to Krifka (1995). On this view, all numerals are assumed to underlyingly designate number concepts but they also involve an additional classifier element that depending on a language can be either overt or covert. There are several variants of this approach but what they all share is that the classifier element takes an integer and returns a counting device equipped with an appropriate measure function (36a) (Scontras 2014, Wagiel 2018). For instance, in the derivation of the phrase *five cats* the arithmetical entity is first turned into a quantifying predicate modifier, which then combines with the NP (36b).

$$(36) \quad \begin{array}{ll} \text{a. } \llbracket \text{CL} \rrbracket = \lambda n_n \lambda P_{\langle e,t \rangle} \lambda x_e [*P(x) \wedge \#(P)(x) = n] \\ \text{b. } \llbracket \text{five cats} \rrbracket = \llbracket [[5 \text{ CL}] \text{ cats}] \rrbracket = \lambda x_e [*CAT(x) \wedge \#(x) = 5] \end{array}$$

In yet another variant of the approach, the classifier semantics turns a numeric value into a predicate (Sudo 2016). The classifier takes a number and yields a cardinal property (type $\langle e, t \rangle$) rather than a predicate modifier. It can also introduce a special presupposition concerning the nature of the referents of the modified noun. The resulting expression then combines with the NP via Predicate Modification.

⁹The exact denotation of the numerical determiner depends on whether the whole sentence gets a collective, distributive or cover reading, i.e., there is a distinct D for each interpretation.

From a theoretical perspective both types of the relationship between the arithmetical and quantifying function, i.e., the derivation from the quantifying meaning to the arithmetical meaning and vice versa, seem plausible. Thus, the question which one is correct is an empirical issue. In the next section, I will discuss morphological evidence useful in determining the direction of the derivation.

4.4 Morphological evidence

Many languages distinguish formally between so-called counting, i.e., arithmetical, and attributive, i.e., quantifying, numerals (Hurford 1998, 2001). For instance, in Mandarin the form *èr* ‘two’ is used in arithmetical environments whereas *liǎng* ‘two’ is an attributive cardinal appearing in the quantifying use. Likewise, the Maltese *tnejn* ‘two’ indicates the arithmetical function while *żewġ* ‘two’ has the quantifying meaning. Though the distinction is typically restricted to a subset of numerals in a language and there are a number of patterns of how the two forms can differ, it is a relatively widespread phenomenon across languages.

Interestingly, the two families of theories discussed above make different predictions regarding the morphological make-up of numerals cross-linguistically. On the assumption that morphology expresses meaning, if the arithmetical function is derived from the quantifying function, see Section 4.2, then we would expect that across languages a pattern with arithmetical numerals being morphologically more complex than quantifying cardinals should be relatively widespread. This is because an additional affix is expected to introduce a shift from a counting device to a number concept. On the other hand, if the quantifying function is derived from the arithmetical one, see Section 4.3, then we would expect an inverse pattern to be common. Namely, quantifying cardinals should include an additional morpheme encoding a shift from number concepts to quantifying modifiers.

It turns out that the cross-linguistically widespread asymmetry attested, e.g., in many obligatory classifier languages, is of the latter type (Wągiel & Caha 2020, Wągiel 2023). For instance, bare numerals in Japanese cannot modify NPs and require an additional element, namely a classifier (37a).¹⁰ However, such an element is illicit in an arithmetical environment like (37b) despite the fact that *ko* being a general classifier could be used to indicate any type of inanimate entity.

¹⁰Typically, classifiers also introduce certain restrictions on the referents of the NP, e.g., being a round or a flat object, a plant, a big animal etc. This aspect of their meaning can be captured by accommodating various presuppositions about the nature of the denoted entities (Sudo 2016).

- (37) a. { *go-no / go-ko-no } ringo
 five-GEN five-CLF-GEN apple
 ‘five apples’
 b. juu waru { go-wa / #go-ko-wa } ni-da.
 ten divided five-TOP five-CLF-TOP two-COP
 ‘Ten divided by five equals two.’ (Japanese)

Therefore, it seems that the approaches that derive the arithmetical semantics from the quantifying one make incorrect predictions with respect to meaning/form correspondences regarding numerals cross-linguistically.¹¹

To conclude, the patterns examined above suggest that the arithmetical meaning of numerals is the basic one and that other uses are derived from it, and thus supporting the approaches discussed in Section 4.3 and challenging those in Section 4.2. In the next section, I will discuss complex numerical expressions in Slavic and suggest how this type of evidence can provide further hints on which of the two ways of relating number concepts and cardinalities discussed so far is on the right track.

5 Complex numerical expressions

While the mainstream semantic research focuses on basic cardinal numerals, there is a growing body of formal literature acknowledging that they are in fact only a subset of various numerical expressions. This fact has been well known in Slavic descriptive traditions. Since Slavic languages have a rich morphology, it is not surprising that one can find abundant formal complexity also in numerals. In this section, I will briefly review recent developments in the study of different types of derivationally and semantically complex numerical expressions.

5.1 Ordinals

The cross-linguistically most widespread type of derived numerical expressions are ordinal numerals, i.e., forms like English *fifth* and Russian *pyatyj* ‘fifth’. Intuitively, ordinals represent position or rank in a sequential order. Modulo suppletion, e.g., *one* ~ *first*, if a language distinguishes morphologically a class of ordinals, they are always derived by an additional affix (Stolz & Veselinova 2013).

Perhaps somewhat surprisingly, the semantics of ordinals has been typically linked to superlatives. Cross-linguistically, the two types of expressions often

¹¹For an analysis of the rare German pattern *ein-s* ~ *ein* ‘one’, see Wągiel & Caha (2020).

share identical or related morphemes (Hurford 1987, Veselinova 1998)¹² and have the same syntax (Barbiers 2007). Furthermore, the acquisition of ordinals seems to be influenced by superlative morphology (Meyer et al. 2018). Even more importantly, ordinals, just like superlatives, give rise to an ambiguity between an absolute and a comparative interpretation (Bhatt 2006, Sharvit 2010). On the absolute reading of (38a), John gave Mary a telescope that is older than all other telescopes. On a comparative reading, however, he might have given her a relatively new telescope than happens to be older than the telescopes other people gave her, i.e., John's gift is compared to other gifts rather than to telescopes in general. Similarly, (38b) can mean either that Mary got a telescope that was older than other telescopes, or that John's telescope was the first Mary has received.

- (38) a. John gave Mary the oldest telescope.
b. John gave Mary the first telescope.

However, ordinals, unlike superlatives, do not give rise to so-called upstairs *de dicto* readings in sentences with intensional operators such as *want* (Bylinina et al. 2015). For instance, (39a) can have a comparative interpretation which makes it true in the scenario below. Yet, that reading is unavailable for (39b).

- (39) SCENARIO: There are many trains throughout the day. John wants to take a train. Any of the trains between 3 pm and 4 pm is fine for him. Similarly, Bill and Steve want to take a train, and they are fine as long as the departure time is between 5 pm and 6 pm and between 7 pm and 8 pm, respectively. These people do not know anything about one another.
- | | |
|---|-------|
| a. John wants to take the earliest train. | TRUE |
| b. John wants to take the first train. | FALSE |

As a result, Bylinina et al. (2015) argue that superlatives and ordinals differ with respect to where they are interpreted. According to that proposal, while superlatives involve movement, ordinals are always interpreted *in situ*.

5.2 Group numerals

One of the key topics in the literature on plurality regards distributive and collective interpretations of sentences including plural DPs (40) (Scha 1981, Link 1983,

¹²Notice that this is also the case in some Slavic languages. For instance, Polish *pierw-szy* and Ukrainian *per-šyj*, both 'first', contain the comparative markers *-szy* and *-šyj* which also appear in superlatives, e.g., *naj-star-szy* and *naj-star-šyj*, both 'the oldest', respectively.

Landman 1989, 2000, Schwarzschild 1996). For instance, (40b) is ambiguous between a reading on which a total of three letters were written, i.e., a distributive interpretation, and a reading on which only one letter was written, i.e., a collective interpretation. The source of the ambiguity has often been located inside the VP (Hoeksema 1983, Link 1984, Schwarzschild 1991; Lasnik 1995: Ch. 7).

- (40) a. The boys wrote a letter.
b. Three boys wrote a letter.

Interestingly, Slavic languages have special numerical expressions which force a collective interpretation of the whole sentence. For instance, consider (41) (Dočekal 2012, 2013).¹³ While (41a) patterns with (40b) in that it is ambiguous between a distributive and a collective reading, (41b) rules out the distributive interpretation, and thus the sole possible reading is that only one letter was written.

- (41) a. *Tři chlapi napsali dopis.* ✓COLL, ✓DISTR
three boys wrote letter.ACC
'Three boys wrote a letter.'
b. *Trojice chlapců napsala dopis.* ✓COLL, # DISTR
three.GRP boys.GEN wrote letter.ACC
'A group of three boys wrote a letter.' (Czech)

Since the difference between the two sentences boils down to the alternation between the basic cardinal numeral *tři* 'three' and the morphologically complex derived form *trojice* 'group of three', the suffix *-ice* was proposed to be a morphological reflex of a semantic operation forcing a collective interpretation, specifically the group-forming operator $\uparrow$ proposed by Landman (1989).¹⁴

Importantly, group numerals cannot be used to refer to number concepts (42). This fact is surprising if the arithmetic meaning were derived from the quantifying one, see Section 4.2. Despite the collective meaning component, NPs with group numerals denote pluralities and one would expect that the same shift that is posited to turn the quantificational meaning of a basic numeral into the name of a number concept should be applicable also in this case, contrary to facts.

¹³In the formal semantic literature, expressions such as *trojice* 'group of three' are also referred to as denumeral group nouns (due to their nominal-like behavior) and collective numerals. However, since the traditional Slavic linguistics uses the latter term for a different type of complex numerals (see Section 5.6), in order to avoid potential confusion I will not use it here.

¹⁴The affix *-oj-* appears only in complex numerical expressions derived from the roots for 2–3 and is usually assumed to have a purely structural function, i.e., it marks non-basic stems.

- (42) a. # Trojice je prvočíslo.
 three.GRP is prime.number
 Intended: ‘Three is prime.’
 b. # Šestice děleno trojicí je dvojice.
 six.GRP divided three.GRP.INS is two.GRP
 Intended: ‘Six divided by three is two.’ (Czech)

Group numerals were identified also in other Slavic languages. In particular, Polish *trójka* ‘group of three’ patterns with (41b) (Wągiel 2015) and similar examples include Slovak *trojica*, Bosnian/Croatian/Montenegrin/Serbian (BCMS) *trojka* and Russian *trojka*, all ‘group of three’.¹⁵ The existence of the discussed phenomenon in Slavic is unexpected since it is typically postulated to hold cross-linguistically that if a language has a scopal quantifier, it is a quantifier forcing a distributive interpretation (Gil 1992).¹⁶ Hence, the Slavic data discussed above compel to revise that generalization. In any case, a systematic theory-driven cross-linguistic research of group numerals has not been carried out so far.

5.3 Both

An interesting feature of many languages is that alongside a regular basic numeral corresponding to English *two* there is also another form corresponding to *both*. In an early GQ account, *both* was analyzed as a quantificational determiner with the same meaning as *the two* (Barwise & Cooper 1981). However, the two expressions are not equivalent since *both* is incompatible with collective predicates (43a) and cannot appear inside partitives (43b) (Ladusaw 1982). The same behavior is also attested in Slavic (44) (Łazarczyk & Pancheva 2009).

- (43) a. { The two / #Both } students are a happy couple.
 b. One of { the two / #both } children sneezed.
 (44) a. # Obe ženščiny prišli vmeste.
 both woman.GEN came together
 Intended: ‘Both women came together.’

¹⁵Russian shows a strong tendency to lexicalize group numerals, e.g., *trojka* is also a word for a carriage drawn by a team of three horses abreast and a special tribunal in the 30s.

¹⁶Hebrew has expressions that resemble Slavic group numerals, e.g., *šlišiya* ‘threesome of’. However, they were reported to differ in that they do not force a collective reading (Gil 1992).

- b. Kto iz ètix { dvoix / *oboix } sdelal to, čto xotel
 who of these.GEN two.GEN both.GEN did this what wanted
 otec?
 father
 ‘Which of these two did what his father wanted?’ (Russian)

The data presented above led to proposals postulating that *both*, unlike basic cardinals, involves an additional distributive component (e.g., Ladusaw 1982, Landman 1989).¹⁷ Interestingly, however, there is yet another respect in which the two differ. Specifically, *both* and its counterparts in other languages cannot be used to refer to a number concept in an arithmetical environment (45).

- (45) a. # Oba to liczba parzysta.
 both that number even
 Intended: ‘Both is an even number.’
 b. # Dziesięć dzielone przez oba równa się pięć.
 ten divided by both.ACC equals REFL five
 Intended: ‘Ten divided by both equals five.’ (Polish)

As group numerals, *both* and its Slavic equivalents pose a challenge for approaches deriving the arithmetical function from the allegedly basic quantifying meaning.

5.4 Taxonomic numerals

A unique property of some Slavic languages is that they have taxonomic numerals. Like group numerals, they are derived by a special suffix which triggers a non-trivial semantic effect. Unlike basic numerals, such expressions do not quantify over object-level entities, i.e., tokens of a type, but rather over subkinds of a particular kind corresponding to the meaning of the modified noun.¹⁸ For instance, let us consider the taxonomic numeral *dvojí* ‘two kinds of’ in Czech (Dočekal 2012, 2013, Grimm & Dočekal 2021). (46a) does not refer to two musical instruments but rather to two types of violin, e.g., a set of classical violins and a set of electric violins. Likewise, (46b) denotes two kinds of the beverage, e.g., white and red wine, irrespective of the amount.¹⁹

¹⁷For recent research on distributive numerals, see Gil (2013), Cable (2014) and Wohlmuth (2019).

¹⁸For an introduction to the kind/object distinction, see, e.g., Krifka et al. (1995).

¹⁹Note that (46) would be more faithfully translated as ‘violins of two kinds’ and ‘wine of two kinds’, but for consistency I follow the convention adopted in Dočekal (2012, 2013).

- (46) a. dvojí housle
two.TAX violin
'two kinds of violin'
- b. dvojí víno
two.TAX wine
'two kinds of wine' (Czech)

Unlike basic numerals, taxonomic numerals specify the domain of quantification to be within the realm of kinds exclusively. Therefore, the suffix *-í* was proposed to introduce a quantificational operation dedicated to counting subkinds, specifically the *KU* (for 'kind unit') operation proposed by Krifka (1995).

Though taxonomic numerals seem to be best preserved in Czech, Polish *dwojaki* and *dwoisty*, Bulgarian *dvojak* and Russian *dvojakij*, all 'twofold', are arguably expressions of a similar type. However, there appear to be certain differences with respect to their distribution and behavior that have not been described sufficiently so far. Crucially, none of the taxonomic numerals can be used to refer to an abstract number concept.

5.5 Aggregate numerals

Another intriguing type of Slavic derivationally complex numerical expressions is sometimes referred to as aggregate numerals, e.g., forms such as Czech *dvoje* 'two collections' (Dočekal 2012, 2013) as well as BCMS *dvoji* (Lučić 2015). Like group numerals, they are derived by a special suffix which triggers a non-trivial semantic effect and lack the arithmetical meaning. While basic numerals simply count atomic individuals in the denotation of the modified NP, aggregate numerals are used to quantify over collections of entities that typically form a particular spatial configuration. For instance, (47a) refers to two aggregates of cards, e.g., two decks of cards and (47b) denotes two pairs of shoes.

- (47) a. dvoje karty
two.AGGR cards
'two sets of cards'
- b. dvoje boty
two.AGGR shoes
'two pairs of shoes' (Czech)

Aggregate numerals are not limited to Slavic. Similar expressions seem to have existed in Latin, e.g., *trēs* 'three' ~ *trīnī* 'three collections' (Ojeda 1997), and Estonian uses pluralized numerals for this purpose, e.g., *kaks* 'two' ~ *kahed* 'two

collections' (Norris 2018). However, it is the Slavic data which was taken as evidence for the relevance of mereotopology in natural language, specifically the significance of topological relations holding between atomic members of a plurality within a part-whole structure.²⁰ In particular, the suffix *-e* in (47) is sensitive to how parts making up a whole are arranged, and thus it was proposed to introduce a semantic operation dedicated to counting clusters, i.e., topologically structured spatial groupings of entities (Grimm & Dočekal 2021).

5.6 Gender-sensitive numerals

All Slavic languages mark grammatical gender on low numerals whereas some of them, e.g., Bulgarian, Polish and Slovak, distinguish between virile (personal masculine) and non-virile forms also in higher numerals (Cinque & Krapova 2007, Pancheva 2018, Wągiel 2023). However, what is probably even more interesting is the existence of complex expressions which I will refer here to as gender-sensitive numerals.²¹ Such expressions introduce non-trivial restrictions with respect to the natural gender of individuals making up a plurality.

Let us consider Polish gender-sensitive numerals such as *troje* 'three (at least one male and at least one female)' (Wągiel 2014, 2015). Unlike virile and non-virile cardinals in (48a)–(48b), they do not show grammatical gender agreement with modified NPs. Instead, they are neuter (48c). And yet they introduce a gender-related effect. While (48a) denotes a plurality of either males or students whose natural gender is not known to the speaker and (48b) refers to three females, (48c) is interpreted as designating a mixed-gender group.

- (48) a. *tych trzech studentów*
 these.v three.v students.GEN.v
 'these three students (male or indeterminate)'
- b. *te trzy studentki*
 these.NV three.NV students.NOM.NV
 'these three female students'
- c. *to troje studentów*
 this.N three.GNDR.N students.GEN.v
 'these three students (at least one male and one female)' (Polish)

²⁰For an introduction to mereotopology, see Grimm 2012: Ch. 4, Wągiel 2018: Ch. 6.

²¹In traditional Slavic linguistics such forms are typically termed "collective numerals". In order to avoid confusion with expressions discussed in Section 5.2, I will not use that term.

Based on the evidence above, numerals such as Polish *troje* were proposed to introduce a presupposition that a designated plurality is heterogeneous and consists of both male and female individuals. Though similar expressions seem to be only attested in BCMS, e.g., *dvoje*, ‘two (one male and one female)’, there are also other types of gender-sensitive numerals in Slavic. For instance, BCMS *dvojica* and Russian *dvoe*, both ‘two (male)’, presuppose a homogeneous group consisting of male individuals only (Kim 2009, Lučić 2015, Khrizman 2020). Interestingly, none of the gender-sensitive numerals can be used to refer to a number concept.

5.7 Indefinite numerals

Another intriguing class of numerical expressions concerns indefinite numerals such as English *several* (Kayne 2007). Similar quantifiers appear in all branches of Slavic, e.g., compare Czech *několik*, Russian *neskol’ko*, Bulgarian *nyakolko* and BCMS *nekoliko*, all ‘several’. Particularly interesting data come from Polish, which distinguishes between two different expressions of this type, specifically *kilka* ‘several, a few’ and *ileś (tam)* ‘some, some amount’. While the former is only compatible with count NPs and designates a cardinality between 3 and 9, the latter also combines with mass NPs and can indicate any quantity (Wągiel 2020b).

In terms of morphosyntax and many aspects of semantics, *several* and its Slavic equivalents pattern with cardinals. However, one crucial respect in which the two differ is that indefinite numerals do not fit contexts calling for number concepts, see (49a) and (50) (Schwarzschild 2002, Wągiel 2020b). Despite the fact that one can easily imagine the intended meaning and paraphrase it using the existential quantifier (49b), this is not what one gets in (49a) and (50).

- (49) a. # Four plus several is less than ten.
 b. There is a number *n* such that four plus *n* is less than ten.
- (50) # Cztery plus { kilka / ileś (tam) } to mniej niż dziesięć.
 four plus several some this less than ten
 Intended: ‘Four plus several/some is less than ten.’ (Polish)

Polish indefinite numerals were analyzed as complex expressions with an incorporated measure function and choice function. The difference between *kilka* and *ileś (tam)* is due to a different type of set the choice function selects a number from and a different kind of built-in measure function (Wągiel 2020b).

5.8 Multipliers

In many languages there is an understudied class of numerical expressions sometimes referred to as multipliers, e.g., English *double*. Though Germanic and Romance languages have borrowed their multipliers from Latin, in Slavic they are morphologically transparent and one can easily notice that they are derived from numerical roots by various affixes, e.g., compare Polish *podwójny*, Russian *dvojnoj* and Czech *dvojítý*, all ‘double’.

Interestingly, multipliers display non-trivial quantificational behavior. Specifically, they do not quantify over whole entities or events, as basic cardinals would, but rather over parts thereof (Wągiel 2018, 2020a). For instance, (51a) denotes a set of singular objects having a complex internal structure. Each of those objects is a crown but crucially it also has two parts that could be considered as having the property of being a crown themselves, e.g., entities such as the ancient Egyptian Pschent or the papal tiara. Similarly, (51b) refers to a murdering event with two victims. But this means that there were two parts of that event each of which could be described as a murder in its own right.

- (51) a. dvostruka kruna
 two.MULT.F crown.N
 ‘double crown’
 b. dvostruko ubistvo
 two.MULT.N murder.N
 ‘double murder’ (BCMS)

The behavior of multipliers stems from the phenomenon of subatomic quantification (Wągiel 2018). According to the proposal, grammar is sensitive to the arrangement of parts within a singular entity. Certain expressions, e.g., proportional partitives and multipliers, can access subatomic part-whole structures and quantify over parts conceptualized as contiguous objects within a whole.

5.9 Multiplicatives

Yet another class of frequent though surprisingly understudied numerical expressions consists of multiplicatives (but see Landman 2006, Donazzan 2013, Kayne 2015). They include adverbial expressions such as English *twice* and *two times* which are primarily used to quantify over events but can also appear in comparatives and equatives. Multiplicatives are attested in all Slavic languages with BCMS *dvaпут* and Czech *dvakrát* ‘two times’ being representative examples.

Intuitively, multiplicatives are part of a broader class of adverbs of quantification which includes frequency adverbs like English *often* (Abeillé et al. 2004, Doetjes 2007). However, close examination reveals interesting differences between them, e.g., in terms of degree modification (Dočekal & Wągiel 2018). For instance, (52a) is ambiguous between the quantified-event and the quantified-degree reading. On the former interpretation, there were two events of increasing the demand and the total value of the increase is unknown whereas on the latter there was a twofold increase in the demand irrespective of the number of events on which it increased. Crucially, (52b) lacks the quantified-degree reading.

- (52) a. Poptávka po dotacích vzrostla dvakrát. ✓EVENT, ✓DEG
 demand after subsidies.LOC increased.PFV twice
 ‘The demand for subsidies increased (by) two times.’
 b. Poptávka po dotacích rostla často. ✓EVENT, #DEG
 demand after subsidies.LOC increased.IPFV often
 ‘The demand for subsidies increased often.’ (Czech)

In addition, while multiplicatives are perfectly fine in comparatives and equatives, frequency adjectives are infelicitous in such constructions (53).

- (53) Petr je { dvakrát / #často } vyšší než Marie.
 Petr is twice often taller than Marie
 ‘Petr is two times taller than Marie.’ (Czech)

Unlike all other types of numerical expressions discussed so far, multiplicatives can be used in arithmetical environments provided that they express multiplication. However, not all multiplicatives can do that. For instance, Polish has two sets of multiplicatives but only one is felicitous in (54) which suggests that the element *razy* ‘times’ is a homonymous form for a multiplicative morpheme used to quantify over events and an expression of an arithmetical relation.

- (54) { Dwa razy / #Dwukrotnie } trzy równa się sześć.
 two times twice three equals REFL six
 ‘Two times three equals six.’ (Polish)

Though multiplicatives are key to understanding event and degree quantification, so far they have received little attention compared to cardinals.

5.10 Frequency numerals

Finally, in many languages there are expressions I will refer to as frequency numerals, i.e., multiplicative adjectives equivalent to English *two-time*. Slavic is no exception here and Polish *dwukrotny*, Czech *dvojnásobný* and Russian *dvukratnyj*, all ‘two-time’, are representative examples of the class in question.

At first sight, it seems that frequency numerals such as *two-time* are expressions of the same type as frequency adjectives like *occasional* and *frequent* (Zimmermann 2003, Schäfer 2007, Gehrke & McNally 2015). However, while *occasional* can have the adverbial reading, *two-time* patterns with *frequent* in that it lacks such an interpretation. For instance, while (55a) means that occasionally a sailor strolled by, both (55b) and (55c) cannot be understood that way.

- | | | | |
|------|----|-----------------------------------|------------|
| (55) | a. | An occasional sailor strolled by. | ✓ADVERBIAL |
| | b. | A frequent sailor strolled by. | #ADVERBIAL |
| | c. | A two-time senator strolled by. | #ADVERBIAL |

Further, frequency numerals fail to combine with sortal nouns and require role nouns denoting socially salient capacities that can be acquired repetitively, e.g., positions with a term and award recipients. That is because they quantify over conventionalized events of acquiring a property, e.g., via a ceremony (56).

- | | | | |
|------|----|---|----------|
| (56) | a. | Jan to dwukrotny { mistrz / #człowiek }. | |
| | | Jan this two-time champion man | |
| | | ‘Jan is a two-time champion.’ | |
| | b. | Jan został { mistrzem / #człowiekiem } dwa razy. | |
| | | Jan became champion.INS man.INS two times | |
| | | ‘Jan became a champion twice.’ | (Polish) |

Let us now summarize what the complex numerical expressions data show us.

5.11 Summary

Complex numerical expressions give rise to various non-trivial semantic effects. The dataset reviewed in this section is far larger and more diverse than typically analyzed in the semantic literature, see Section 3. And yet, there is an intuition that a general theory of numerals should cover all of the examined cases. Therefore, I believe that the relevance of the discussed Slavic data is twofold.

First, morphological transparency of the discussed forms clearly shows that (unless suppletion is involved) all complex numerical expressions share a common component, i.e., a numerical root designating a number. That number is

somehow employed in counting but what exactly is counted may vary depending on the type of numeral, e.g., whole objects in the case of basic cardinals, parts in the case of multipliers and subkinds in the case of taxonomic numerals. Hence, a theory of numerals should aim at providing a general mechanism for capturing not only the behavior of basic numerals but also other numerical expressions.

Second, despite sharing an element designating a number, none of the complex numerical expressions can be used to refer to number concepts, i.e., the arithmetical function is never available except for basic numerals. This fact is problematic for the approaches deriving the arithmetical meaning from the quantifying one, see Section 4.2. The reason is that they shift a counting-device semantics to a number name. But if so, why can that shift not be also applied in other types of counting expressions, e.g., taxonomic or indefinite numerals, or *both*? Semantically, they do not seem to differ radically from basic cardinals (they all count something) to justify the unavailability of the shifting. If, on the other hand, the arithmetical meaning is basic, one expects to derive various quantificational flavors from it. Therefore, it would not be surprising that only basic numerals can have the arithmetical function while other types of numerical expressions lack it.

Based on the two conclusions reached above, in the next section I will propose an analysis of yet another type of numerical expressions, namely label numerals.

6 Case study: Numerals as labels

Let us now return to the label use of numerals introduced in (1f), repeated here as (57), and discuss how it relates to the quantifying and arithmetical function.

(57) Player number five twisted her ankle.

So far, the label meaning has received little attention in the literature (see Hurford 1987: 167–168; Wiese 2003: 37–42; Bultinck 2005: 119–129). I will demonstrate that not only is this function conceptually different from other functions discussed in Section 2 but also that the distinction is reflected in grammar. I will argue that the label function is derived from the arithmetical meaning and I will propose a morpho-semantic account that allows for deriving both the form and the meaning of dedicated label numerals in Slavic. The proposed analysis will combine standard compositional semantics with a Nanosyntax model of morphology.

6.1 Label uses are distinct

Label numerals provide means for identification of an object via its association with a unique number, which allows for the recognition of that object among other entities. This mechanism is sometimes referred to as NOMINAL NUMBER ASSIGNMENT (Wiese 2003: 37–38). A set of numbers are consistent labels as long as the numerals corresponding to distinct numbers in that set are uniquely used for each distinct object. This can be achieved by one-to-one mapping between a set of labeled entities and a given set of numbers. Importantly, the actual values of the numbers represented by label numerals are irrelevant since they do not indicate quantity, order or any other type of measurement. Furthermore, the set of labels can change over time, e.g., it can grow as new entities are labeled. The only thing that matters is that at a given time every labeled entity is associated with a unique number.

It is clear that from a conceptual point of view the label function differs from the quantifying and the arithmetical meaning. Importantly, however, the distinction is also encoded in grammar. First, numerals used as labels do not allow for modification by numeral modifiers and prequantifiers (58a). Moreover, label constructions do not give rise to scalar implicatures. For instance, in the context of (58b), taking tram number six would not guarantee getting to the university.

- (58) a. # Tram { more than / at least / all } number five has left.
 b. You must take tram number five to get to the university. #AT LEAST

The data in (58) differ from the quantifying function and pattern with the properties of the arithmetical use of numerals discussed in Section 2.3. However, these two functions can be distinguished linguistically as well. For instance, English constructions with numerals used as labels differ from other count NPs in that they are ungrammatical with the indefinite article (59a). On the other hand, names of number concepts cannot appear without the definite article (59b) (for a detailed discussion of English phrases such as *number five*, see Bultinck 2005: 124–129).

- (59) a. Kim took (*a) tram number five.
 b. * (The) number five is odd.

Above, I argued that the label meaning of numerals is both conceptually and linguistically different from the quantifying and the arithmetical function. In the following two sections, I will discuss further evidence for the relevance of the label function as well as its relationship with the other two discussed meanings.

6.2 Evidence from co-lexicalizations

In many languages, an additional expression can be optionally used alongside the numeral in order to indicate its quantifying, arithmetical or label use. Sometimes, the same lexical item can introduce all of the functions above. For instance, English employs a single noun, namely *number*, to indicate cardinality, to designate a mathematical object and to signal that an entity is identified via association with a relevant integer (60).

- (60) a. The cats are five in number.
b. The number five is odd.
c. Tram number five left.

While English shows full co-lexicalization, i.e., the noun *number* can be used to indicate all three functions, more often lexicons develop a different word for at least one of the functions in question. For instance, Bulgarian exhibits a pattern with no co-lexicalization, i.e., each of the discussed functions is signalled by a morphologically distinct expression, specifically *broj* is for quantification, *číslo* is for arithmetic and *nomer* is for the label function, all ‘number’.

Though ways in which meanings and forms correspond to each other vary, it seems that they do not vary in an unrestricted manner. The co-lexicalization patterns found in Slavic are summarized in Table 1. Interestingly, the pattern that is not attested in the sample is a case of non-contiguous co-lexicalization where the quantifying and label function are expressed with the same form while the arithmetical function is expressed by a different lexical item. Perhaps surprisingly, this result resembles the well-known *ABA principle (Bobaljik 2012).

Table 1: Co-lexicalization of ‘number’

LANGUAGE	QUANTIFYING	ARITHMETICAL	LABEL
BCMS	A broj	A broj	A broj
Polish	A liczba	A liczba	B numer
Slovak	A počet	B číslo	B číslo
Bulgarian	A broj	B číslo	C nomer
unattested	A	B	A

Examining permissible patterns of co-lexicalization is rarely considered evidence in formal semantics. However, I believe that looking at how certain parts

of the lexicon are structured can be instructive. Arguably, it can reveal that some meanings have more in common than others and in some cases that certain categories, though theoretically possible, are empirically unrealized.²² Consequently, one can gain valuable insights regarding what those meanings are.

On the assumption that co-lexicalization does in fact reflect the relationship between concepts, the data in Table 1 suggest that while the quantifying and the label meaning are unrelated, they both seem to stem from the arithmetical function. The reason is that if both the quantifying and the label meaning are derived from the arithmetical meaning, the attested co-lexicalization patterns are unsurprising since the unrelated derived functions are not expected to be expressed by a single term that is different from the one expressing the underlying meaning (ABA). On the other hand, if the quantifying meaning were basic, then one would expect to find the ABA pattern instead of the Czech and Slovak ABB.

Whether a language which has a term for the quantifying and the label function to the exclusion of the arithmetical function can be found is an empirical issue. So far, the Slavic co-lexicalization data suggest a strong relationship between the label and the arithmetical meaning and no relationship between the label and the quantifying meaning. Let us now move to Slavic label numerals.

6.3 Derived label numerals

An interesting property of Slavic languages is that they have specialized label numerals. Such dedicated expressions are derived with the suffixes *-ka* and *-ica*. Table 2 gives the label forms for the numerals corresponding to 5 across Slavic.

Table 2: Cardinal and label ‘five’ in Slavic

LANGUAGE	NUMBER	CARDINAL	LABEL
Czech	5	pět	pětka
Polish	5	pięć	piątka
Russian	5	pjat’	pjaterka
Slovenian	5	pet	petka/petica
BCMS	5	pet	petica

Though there are a number of interesting differences in their productivity and distribution, e.g., Polish derives label forms with *-ka* up to 999 while Russian has

²²This way of viewing data has been successful in lexicology (see Hage 1997 on kinship terms).

them only for 2–10, 20 and 30, all Slavic label numerals are morphosyntactically nominal expressions used to refer to objects that can be identified via nominal number assignment. Though very often they appear bare, the noun specifying what entity is labeled can optionally precede the label numeral (61). In the absence of the noun this information is typically provided by the context.

- (61) (Tramvaj) pět-ka vyjede na novou trať v listopadu.
 tram five-LBL will.go.out on new.ACC track.ACC in November.LOC
 ‘Tram number five will head out on the new track in November.’ (Czech)

Similar to classifier constructions and other complex numerical expressions, recall Section 4.4 and Section 5, Slavic label numerals exhibit uniform behavior in that they cannot be felicitously used in mathematical statements like (62). This, in turn, indicates that their semantics is more intricate than the arithmetical meaning.

- (62) a. Dva-krát pět se rovná deset.
 two-times five REFL equals ten
 ‘Two times five equals ten.’
 b. # Dvoj-ka krát pět-ka se rovná desít-ka.
 five-LBL times two-LBL equals ten-LBL
 Intended: ‘Two times five equals ten.’ (Czech)

Thus, Slavic label numerals are both morphologically and semantically complex. Let us now propose an account that will capture their morpho-semantics and the relationship between the arithmetical meaning, quantification and labeling.

6.4 Analysis

In Section 2 and 4, we saw the theoretical relevance of the fact that numerals are often ambiguous. In order to account for the relationship between the three uses of basic numerals this paper focuses on, i.e., their arithmetical, quantifying and label function, I develop a morpho-semantic system combining a standard inventory of formal semantics with the Nanosyntax model of morphology (see Wągiel & Caha 2020). The core of the proposal is that the quantifying and the label function are both derived from the arithmetical meaning which is taken to be basic.

I propose the ingredients in (63) as a set of morpho-semantic components that numeral expressions conveying the relevant functions are formed of. Specifically, I postulate the bottom-most NUM_n (for ‘number’) layer along with two additional heads, namely CL (for ‘classifier’) and LBL (for ‘label’).

- (63) a. $\llbracket \text{NUMP}_n \rrbracket_n = n$
b. $\llbracket \text{CL} \rrbracket_{\langle n, \langle \langle e, t \rangle, \langle e, t \rangle \rangle \rangle} = \lambda n \lambda P \lambda x [*P(x) \wedge \#(P)(x) = n]$
c. $\llbracket \text{LBL} \rrbracket_{\langle n, \langle \langle e, t \rangle, e \rangle \rangle} = \lambda n \lambda P t x [P(x) \wedge \text{LABEL}(n, x)]$

The meaning of NUMP_n is simply an abstract number concept (type n) (63a). The subscript indicates that NUMP_n is lexically encoded. Though for the purpose of this paper I assume that it is the basic structural component of each numeral, I do not claim that it is primitive. It might very well be further decomposed into a number of features.²³ On the other hand, the components CL and LBL are interpreted as shifts from n to a more complex type.

I will assume here that CL is interpreted as a function from a number concept to a predicate modifier devised for counting individuals (type $\langle n, \langle \langle e, t \rangle, \langle e, t \rangle \rangle \rangle$) (63b).²⁴ CL takes a number and yields an expression with the built-in pluralization operation $*$ and measure function $\#(P)$, see Section 3.3. In cases such as English *five* or Czech *pět* ‘five’, CL is already incorporated in the structure of a numeral but in some other languages it is encoded in an additional morpheme, e.g., a classifier.

On the other hand, LBL is a labeling device (type $\langle n, \langle \langle e, t \rangle, e \rangle \rangle$) which selects a number and returns a function from properties to unique individuals such that they have the relevant property and are labeled with the corresponding number by the LABEL operation (63c). LBL is typically introduced by a free morpheme like *number* or a specialized affix like the Czech suffix *-ka*, as discussed in Section 6.2–6.3.

As a result of combining the components in (63), we obtain the structures in Figure 1 and 2. Figure 1 represents the meaning of the quantifying function of a numeral corresponding to 5 whereas Figure 2 depicts the parallel label function.

When a quantifying numeral, e.g., Czech *pět* ‘five’, and a label expression, e.g., Czech *číslo pět* ‘number five’, combine with an NP, we obtain denotations as in (64). In particular, (64a) denotes a set of pluralities of players such that each plurality in that set consists of five players. On the other hand, (64b) refers to the contextually unique player identified via their association with the number 5.

- (64) a. $\llbracket \text{pět hráčů} \rrbracket_{\langle e, t \rangle} = \lambda x [* \text{PLAYER}(x) \wedge \#(\text{PLAYER})(x) = 5]$
b. $\llbracket \text{hráč číslo pět} \rrbracket_e = \iota x [\text{PLAYER}(x) \wedge \text{LABEL}(5, x)]$

²³This is indicated by the triangle in Figure 1 and 2. For arguments why one would want to further decompose NUMP_n , see Wagiel & Caha (2020).

²⁴For the purpose of this paper, it is not important what the exact interpretation of CL is. The proposed system is compatible with other theories of quantifying numerals discussed in Section 3 as long as it is possible to shift the primitive type n to a type for the quantifying function.

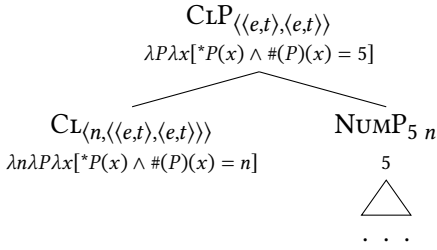


Figure 1: The quantifying function

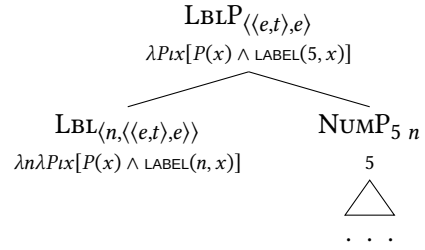


Figure 2: The label function

But how to account for the fact that *pět* ‘five’ can be used to express both the quantifying and the arithmetical meaning? This ambiguity can be captured as a consequence of a core property of late insertion, namely that lexical entries are not tailor-made for one specific use.²⁵ In Nanosyntax, a lexical entry pairs a well-formed syntactic structure with phonology. For instance, let us assign to the numeral *pět* the lexical entry in Figure 3. When syntax builds a structure interpreted as the quantifying meaning, as in Figure 1, the entry in Figure 3 can be used to lexicalize it. Consequently, the structure or, more precisely, its top-most phrasal node is pronounced by /*pjet*/, as indicated by the big circle in Figure 4.

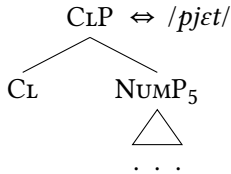


Figure 3: Lexical entry

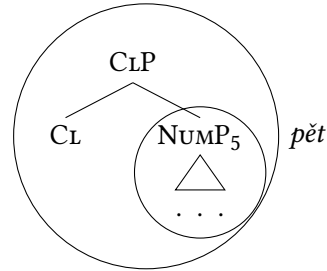


Figure 4: Lexicalizations

Crucially, however, when syntax builds a structure interpreted as the arithmetical meaning, i.e., the sole NUMP₅, it can also be pronounced by Figure 3, as indicated by the small circle in Figure 4. This is due to the Superset Principle, defined in (65) (Starke 2009), which states that a lexical entry can be used to pronounce a particular structure iff it contains that structure as a sub-part (either proper or improper). As a result, the Czech numeral *pět* ‘five’ is ambiguous between the quantifying and the arithmetical meaning since both circled structures in Figure 4 are contained in the tree in Figure 3.

²⁵For an introduction to late insertion models such as Distributed Morphology and Nanosyntax, see, e.g., Bobaljik (2017) and Baunaz & Lander (2018), respectively.

(65) The Superset Principle (Starke 2009)

A lexically stored tree matches a syntactic node iff the lexically stored tree contains the syntactic node.

Let us now consider how the label meaning arises. For this purpose, I will demonstrate how specialized label numerals such as Czech *pětka* ‘number five’ are derived. Since *pět* ‘five’ is stored as Figure 3, and thus cannot express the label meaning, the LBL component needs to be spelled out by a separate entry, e.g., as the suffix *-ka* in Figure 5. Notice that the semantics of LBL is not compatible with the quantifying meaning and requires type *n* as its input (63c). Thanks to the Superset Principle (65), *pět* can also pronounce the arithmetical meaning, i.e., the sole NUMP_5 , as indicated by the small circle in Figure 4. However, in order for this structure to apply at the LBLP node, NUMP_5 must move from its base-position. In Figure 6, the lower copy of the displaced element is shaded.²⁶ After the NUMP_5 is moved, Figure 5 matches the lower LBLP, as shown in Figure 7, where the label suffix *-ka* is inserted at the relevant node.

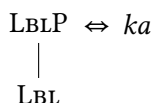


Figure 5: Czech label suffix

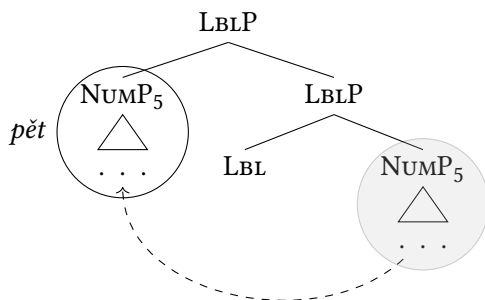


Figure 6: Movement

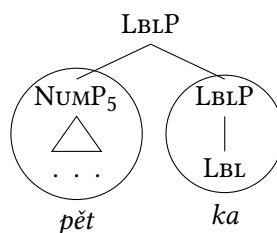


Figure 7: Spellout

The proposed analysis has several advantages. First of all, it provides a unified approach that captures the relationship between the arithmetical, the quantifying and the label meaning. Second, it explains when and how a numeral can be

²⁶In Nanosyntax, the movement in Figure 6 is driven by the need to lexicalize the phrasal LBLP node by the lexical entry in Figure 5. For a detailed explanation of how precisely the movement in Figure 6 is triggered, see Baunaz & Lander (2018) and Caha et al. (2019).

ambiguous between different meanings. Finally, it gives a detailed account for the meaning/form correspondences both in cardinal and in label numerals.

7 Conclusion

In this paper, I explored the semantics of numerals in Slavic and beyond. I discussed various functions that basic numerals can have as well as different approaches to capturing the relationships between these functions. Furthermore, I examined many types of complex numerical expressions that only recently attracted attention in the literature. I also provided novel evidence in favor of the view suggesting that the arithmetical meaning is basic while other functions are derived from it. Finally, I explored the so-far understudied label function and proposed a morpho-semantic analysis of Slavic derived label numerals that could be further extended to other types of complex numerical expressions.

Abbreviations

ACC	accusative	LOC	locative
AGGR	aggregate numeral	MULT	multiplier
CLF	classifier	N	neuter
COP	copula	NBR	number-denoting numeral
F	feminine	NOM	nominative
GEN	genitive	NV	non-virile
GNDR	gender-sensitive numeral	PFV	perfective
GRP	group numeral	REFL	reflexive pronoun
INS	instrumental	TAX	taxonomic numeral
IPFV	imperfective	TOP	topic
LBL	label numeral	V	virile

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